AI-Based Multi-Constraint Delivery Route Optimizer

SRS, System Design & Methodology Review

Design Review Presentation

College of Engineering Chengannur
Dept. of Computer Engineering
CSD.415 Project Phase 1

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Outline

1. Proposed System
2. SRS
3. System Design
4. Modules
5. Methodology
6. Results
7. Future Scope
8. Conclusion

Proposed System — Recap

- A web application that generates delivery routes optimized for vehicle capacity, delivery time windows, and delivery priority, using Ant Colony Optimization (ACO).
- Dynamically re-optimizes routes when a road closure is reported.
- A Genetic Algorithm independently solves the same problem for comparison.
- An analytics dashboard summarizes optimization benefit across completed deliveries.

Functional Requirements

1. The system shall allow users to add delivery stops with weight, time window, and priority level.
2. The system shall validate total load against vehicle capacity before optimization.
3. The system shall generate an optimized route using ACO, respecting capacity, time-window, and priority constraints.
4. The system shall allow an admin to mark a road segment as closed and trigger live re-optimization.
5. The system shall independently solve the same routing problem using a Genetic Algorithm for comparison.
6. The system shall display convergence graphs for both algorithms.
7. The system shall display the optimized route on an interactive map.
8. The system shall maintain an analytics dashboard summarizing distance saved, success rate, and average delivery time.
9. The system shall store route history and delivery records in a database.
10. The system shall allow users to edit or remove delivery stops before optimization.

Non-Functional Requirements

- **Performance:** Route optimization should complete within a few seconds for up to 50 delivery stops.
- **Usability:** Simple, intuitive web interface requiring no user training.
- **Reliability:** The system should handle invalid inputs (e.g., overweight loads) gracefully, without crashing.
- **Scalability:** The architecture should support an increasing number of stops and vehicles in future extensions.
- **Maintainability:** Modular codebase separating algorithm logic, API, and presentation layers.
- **Portability:** Fully web-based, accessible from any modern browser, no installation required.
- **Security:** Input validation and sanitization for all user-submitted data.

Hardware / Software Requirements

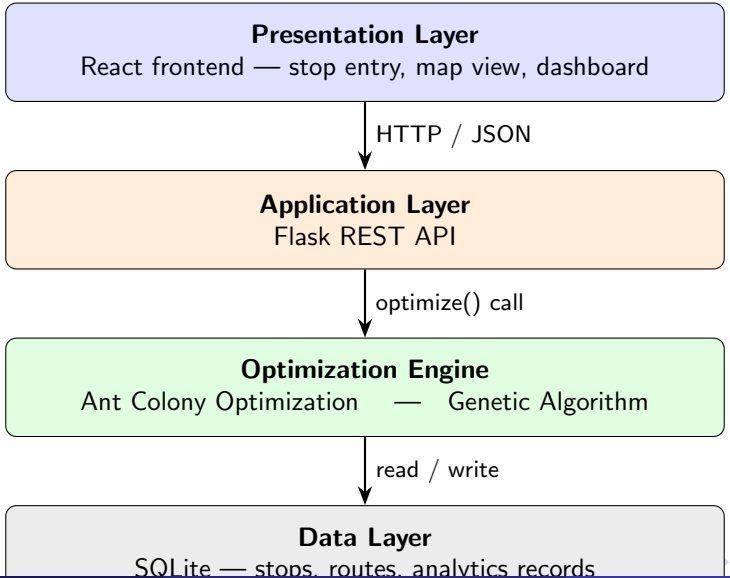
Hardware

- Development machine: Intel i5 (or equivalent) or higher
- Minimum 8 GB RAM
- Standard broadband internet connection
- No specialized hardware required for deployment

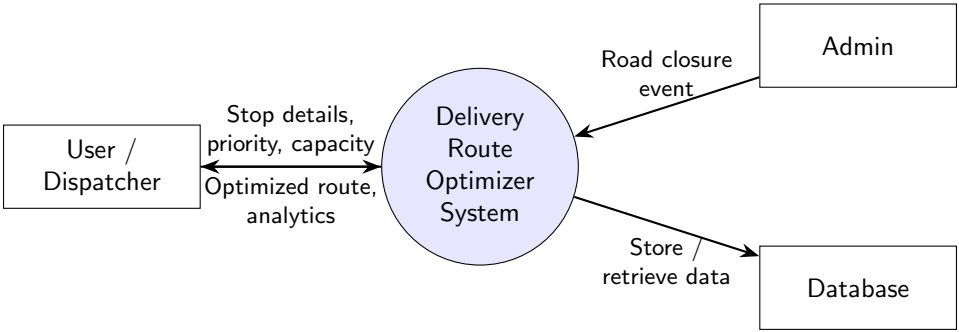
Software

- Python 3.x, Flask (backend API)
- SQLite (database)
- Node.js, React (frontend)
- react-leaflet (map), recharts (charts)
- VS Code, Git & GitHub
- Overleaf / LaTeX (documentation)

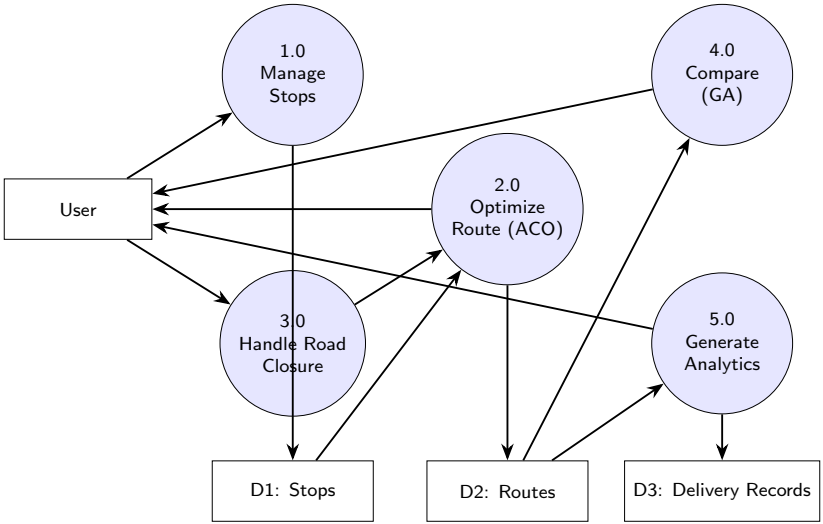
System Architecture



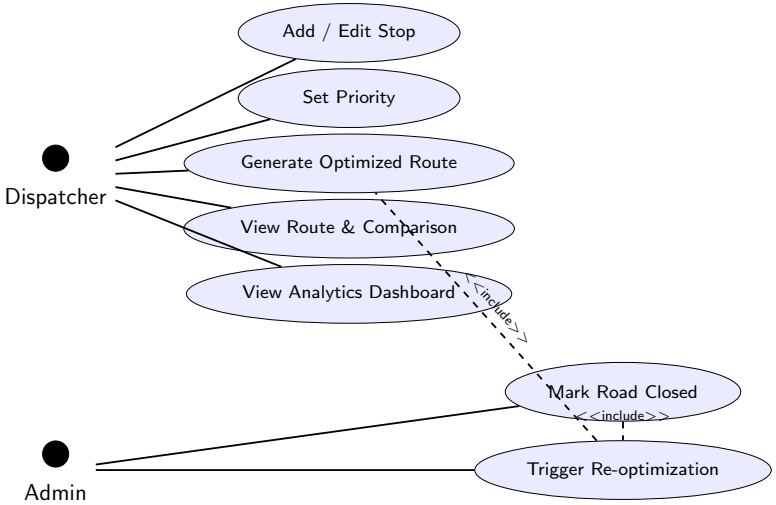
Data Flow Diagram — Context Diagram (Level 0)



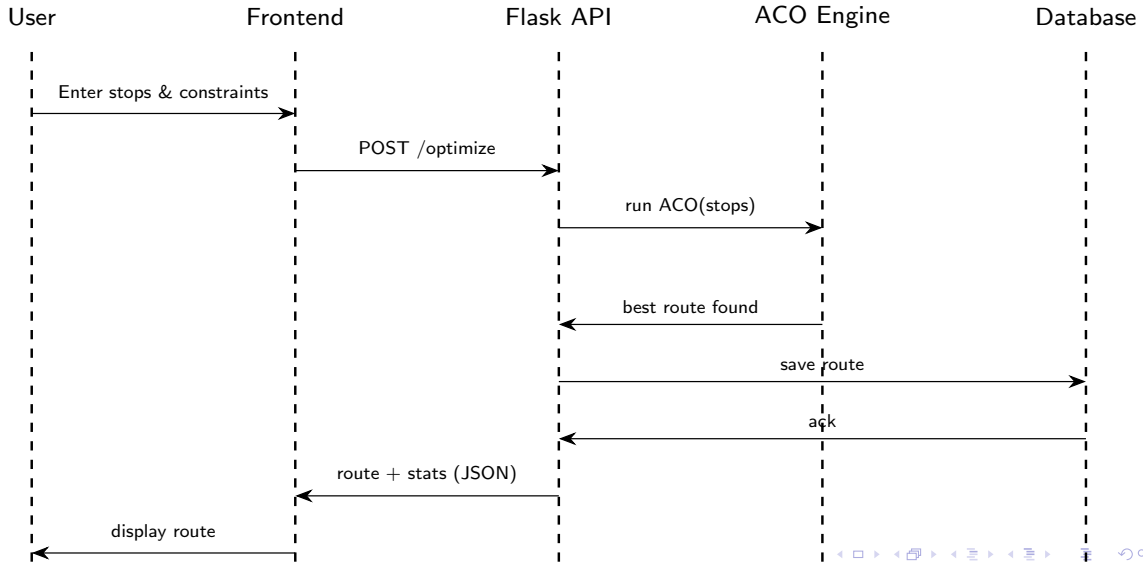
Data Flow Diagram — Level 1



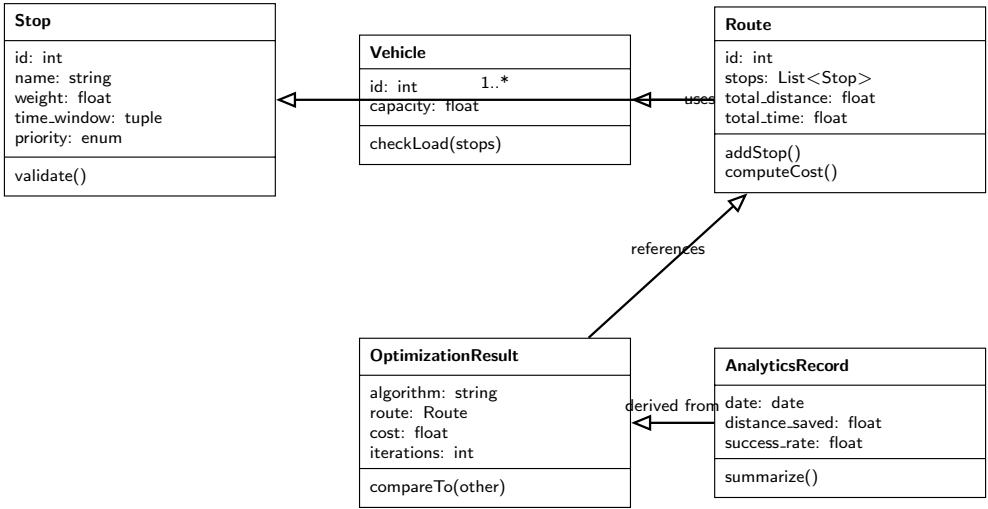
Use Case Diagram



Sequence Diagram — Generate Optimized Route



Class Diagram



System Modules

1. **Stop Management Module** — add, edit, remove delivery stops and their constraints.
2. **Route Optimization Module (ACO)** — core route-finding engine.
3. **Comparison Module (GA)** — independent Genetic Algorithm solver for benchmarking.
4. **Road Closure / Dynamic Re-routing Module** — handles live disruption and re-optimization.
5. **Analytics Dashboard Module** — aggregates and displays performance metrics.
6. **Map Visualization Module** — renders routes and closures on an interactive map.
7. **Database / Persistence Module** — stores stops, routes, and delivery records.

Methodology — Recap

- **ACO:** Simulated ants build candidate routes probabilistically (pheromone strength + distance + priority), reinforcing shorter, valid routes over iterations.
- **GA:** Candidate routes evolve via selection, crossover, and mutation across generations, using the same constraint-aware cost function.
- **Development approach:** Iterative development aligned with the project phase schedule; Git for version control; Flask REST API backend with a React frontend.

Expected Results

Since the project is currently at the design stage, the figures below represent target benchmarks derived from preliminary simulation and the reviewed literature, to be validated once implementation is complete.

Metric	ACO	Genetic Algorithm
Best route distance (sample instance)	18.4 km	19.1 km
Iterations/generations to converge	140	210
Re-optimization time after closure	< 1 sec	–

Target overall system benchmarks: $\sim$**18% distance reduction** vs. an unoptimized baseline route, and $>$95% on-time delivery success rate in simulation.

Future Scope

- Multi-depot optimization — automatically select the best depot per delivery.
- Integration with real-time GPS and traffic data instead of simulated closures.
- Multi-vehicle fleet assignment and scheduling.
- Driver-facing mobile app for live route updates.
- Machine-learning-based demand forecasting to anticipate delivery volume.

Conclusion

- The system design addresses a realistic, multi-constraint delivery routing problem — capacity, time windows, priority, and live disruptions — going well beyond a basic shortest-path solver.
- The architecture cleanly separates presentation, application, optimization, and data layers, supporting maintainability and future extension.
- The SRS, DFDs, use case, sequence, and class diagrams presented here define a clear, implementable scope for the next project phase.

References I



A. Maroof, B. Ayvaz, and K. Naeem, "Logistics Optimization Using Hybrid Genetic Algorithm (HGA): A Solution to the Vehicle Routing Problem With Time Windows (VRPTW)," *IEEE Access*, vol. 12, pp. 36974–36989, 2024. (IEEE Access, open access)



L. Wu, Z. He, Y. Chen, D. Wu, and J. Cui, "Brainstorming-Based Ant Colony Optimization for Vehicle Routing With Soft Time Windows," *IEEE Access*, vol. 7, pp. 19643–19652, 2019. (IEEE Access, open access)



M. Liu, Q. Song, Q. Zhao, L. Li, Z. Yang, and Y. Zhang, "A Hybrid BSO-ACO for Dynamic Vehicle Routing Problem on Real-World Road Networks," *IEEE Access*, vol. 10, pp. 118302–118312, 2022. (IEEE Access, open access)

References II



Z. Hao, Y. Wang, and X. Yang, "Every Second Counts: A Comprehensive Review of Route Optimization and Priority Control for Urban Emergency Vehicles," *Sustainability*, vol. 16, no. 7, article 2917, 2024. (MDPI, open access)



N. Insani, S. Taheri, and M. Abdollahian, "A Mathematical Model for Integrated Disaster Relief Operations in Early-Stage Flood Scenarios," *Mathematics*, vol. 12, no. 13, article 1978, 2024. (MDPI, open access)



X. Liu, Y.-L. Chen, L. Y. Por, and C. S. Ku, "A Systematic Literature Review of Vehicle Routing Problems with Time Windows," *Sustainability*, vol. 15, no. 15, article 12004, 2023. (MDPI, open access)



X. Ma and C. Liu, "Improved Ant Colony Algorithm for the Split Delivery Vehicle Routing Problem," *Applied Sciences*, vol. 14, no. 12, article 5090, 2024. (MDPI, open access)